241401084

CONTIGUOUS MEMORY ALLOCATION

1)first fit

user123@localhost:~/memory

```
[user123@localhost ~]$ cd memory
[user123@localhost memory]$ vi firstFit.c
[user123@localhost memory]$
```

user123@localhost:~/memory — /usr/bin/vim firstFit.c

```
#include <stdio.h>

int main(){
        int n, m;

        printf("Enter number of memory blocks:\n");
        scanf("%d",&m);

        int blockSize[m], remainingSize[m];

        printf("Enter size of each block:\n");
        for(int i=0; i<m; i++){
                scanf("%d", &blockSize[i]);
                remainingSize[i] = blockSize[i];
        }

        printf("Enter number of processes:\n");
        scanf("%d",&n);

        int processSize[n];

        printf("Enter size of each process:\n");
        for(int i=0; i<n; i++){
                scanf("%d",&processSize[i]);
        }

        for(int i=0; i<n; i++){
                int allocated = 0;

                for(int j=0; j<m; j++){
                        if(remainingSize[j]>=processSize[i]){
                                int fragment = remainingSize[j] - processSize[i];

                                printf("Process %d of size %d is allocated to Block %d of size %d with Fragment %d\n", i+1, processSize[i], j+1, blockSize[j],
 fragment);

                                remainingSize[j] -= processSize[i];

                                allocated = 1;
                                break;
                        }
                }
```

```
                if(!allocated){
                        printf("Process %d of size %d is not allocated\n", i+1, processSize[i]);
                }
        }

        return 0;
}
-- INSERT --                                                                    49,1          Bot
```

```
[user123@localhost ~]$ cd memory
[user123@localhost memory]$ vi firstFit.c
[user123@localhost memory]$ gcc firstFit.c
[user123@localhost memory]$ ./a.out
Enter number of memory blocks:
5
Enter size of each block:
100
500
200
300
600
Enter number of processes:
4
Enter size of each process:
212
417
112
426
Process 1 of size 212 is allocated to Block 2 of size 500 with Fragment 288
Process 2 of size 417 is allocated to Block 5 of size 600 with Fragment 183
Process 3 of size 112 is allocated to Block 2 of size 500 with Fragment 176
Process 4 of size 426 is not allocated
[user123@localhost memory]$
```

2)best fit

```
[user123@localhost ~]$ cd memory
[user123@localhost memory]$ vi bestFit.c
[user123@localhost memory]$
```

```c
#include <stdio.h>

int main(){
        int m, n;

        printf("Enter number of memory blocks:\n");
        scanf("%d",&m);

        int blockSize[m], remaining[m];

        printf("Enter size of each block:\n");
        for(int i=0; i<m; i++){
                scanf("%d", &blockSize[i]);
                remaining[i] = blockSize[i];
        }

        printf("Enter number of processes:\n");
        scanf("%d",&n);

        int processSize[n];

        printf("Enter size of each process:\n");
        for(int i=0; i<n; i++){
                scanf("%d", &processSize[i]);
        }

        for(int i=0; i<n; i++){
                int bestIndex = -1;

                for(int j=0; j<m; j++){
                        if(remaining[j] >= processSize[i]){
                                if(bestIndex==-1 || remaining[j] < remaining[bestIndex]){
                                        bestIndex = j;
                                }
                        }
                }

                if(bestIndex != -1){
                        int fragment = remaining[bestIndex] - processSize[i];
                        printf("Process %d of size %d is allocated to Block %d of size %d with Fragment %d\n", i+1, processSize[i], bestIndex+1, blockSize[bestIndex], fragment);
                        remaining[bestIndex] -= processSize[i];
                } else {
                        printf("Process %d of size %d is not allocated\n", i+1, processSize[i]);
                }
        }
        return 0;
}
-- INSERT --                                                        48,1          Bot
```

user123@localhost:~/memory

```
[user123@localhost ~]$ cd memory
[user123@localhost memory]$ vi bestFit.c
[user123@localhost memory]$ gcc bestFit.c
[user123@localhost memory]$ ./a.out
Enter number of memory blocks:
5
Enter size of each block:
100
500
200
300
600
Enter number of processes:
4
Enter size of each process:
212
417
112
426
Process 1 of size 212 is allocated to Block 4 of size 300 with Fragment 88
Process 2 of size 417 is allocated to Block 2 of size 500 with Fragment 83
Process 3 of size 112 is allocated to Block 3 of size 200 with Fragment 88
Process 4 of size 426 is allocated to Block 5 of size 600 with Fragment 174
[user123@localhost memory]$
```